

Bharath Kumar Rajesh

AI ENGINEER

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AI Engineer who ships production agentic systems and the reliability tooling that makes them trustworthy: multi-agent orchestration, RAG pipelines, and LLM evaluation, observability, and governance on Claude (Amazon Bedrock) and AWS. M.S. Computer Science, Pace University (GPA 3.86). Anthropic and AWS certified, with two peer-reviewed publications and a public open-source track record across agent tooling, MCP, and CNCF projects.

TECHNICAL SKILLS

AI / Agentic: LangChain, LangGraph, Multi-Agent Orchestration (ReAct, Tool-Use, Memory), MCP (Model Context Protocol), RAG Pipelines, Embeddings, Vector Databases (ChromaDB, Pinecone, FAISS, Weaviate), Prompt Engineering, LoRA Fine-Tuning

Models & Evaluation: Claude (Amazon Bedrock), OpenAI API, LLM Evaluation Pipelines, Automated Graders, LLM-as-Judge, Drift Monitoring, OpenTelemetry, Langfuse, Prometheus, Grafana

Languages: Python (Advanced), Go, TypeScript, JavaScript, Java, SQL, C++, R, Bash

Backend & Full-Stack: FastAPI, Flask, Spring Boot, Node.js, Express, gRPC, GraphQL, REST, WebSockets, Microservices, React, Next.js, Tailwind, Redux

Cloud & DevOps: AWS (EC2, S3, Lambda, SageMaker, ECS, RDS, IAM, Kinesis), Azure (App Service, Functions), Kubernetes, Docker, Terraform, GitHub Actions, CI/CD, Helm, Linux

Data: PostgreSQL, MongoDB, Redis, Elasticsearch, MySQL, DuckDB, Pandas, NumPy, Apache Spark, Apache Airflow, AWS Kinesis, ETL

ML & Frameworks: PyTorch, TensorFlow, scikit-learn, MLflow, XGBoost, OpenCV, NLP, ONNX

AI Dev Tools: Claude Code, Cursor, GitHub Copilot

EXPERIENCE

Graduate Assistant, Applied AI Engineering

March 2025 to Present

Pace University, Seidenberg School of Computer Science

New York, NY

- Built and deployed a multi-agent chatbot platform serving 10,000+ daily users at 99.5% uptime, using LangGraph orchestration, Claude on Amazon Bedrock, and custom MCP tool servers with department-specific routing.
- Engineered RAG pipelines indexing 50,000+ documents, reaching 92% MRR@5 and cutting retrieval latency 60%.
- Built LLM observability dashboards tracking latency, token throughput, cost, drift, and eval scores across production workflows.
- Shipped 12 zero-regression releases through automated CI/CD with 85%+ test coverage; developed daily with Claude Code and Cursor.

Open Source Contributor, Artie Labs (CDC Streaming Platform)

March 2026 to April 2026

artie-labs/transfer

Remote

- Contributed three pull requests to an active Go change-data-capture platform (#1731 contribution guide, #1732 destination client, #1733 source consumer).
- Implemented an Elasticsearch destination client in Go using the `_bulk` API, with type mapping across `STRING`, `TEXT`, `INTEGER`, `BIGINT`, `FLOAT`, `EDecimal`, `TIME`, and `Bytes`, plus composite-key URL-encoding to prevent collisions; resolved nine bugs across review rounds.
- Implemented an AWS Kinesis source consumer in Go (aws-sdk-go-v2) with a shard-to-index map, monotonic offset counter, and per-shard rate-limit handling; resolved three concurrency bugs with all tests passing.

Software Development Intern

September 2023 to May 2024

Let's Be The Change

Bangalore, India

- Owned a full-stack Java/Spring Boot and React platform serving 1,000+ users end to end.
- Re-architected SQL query patterns and introduced Redis caching, dropping API P99 latency 80% (2,000ms to 400ms).
- Integrated Stripe (payments), SendGrid (email), and Twilio (messaging) APIs with zero post-launch incidents.
- Scoped and shipped an A/B experiment on recommendation logic in under three weeks; the winning variant delivered a 20% retention lift and 35% reduction in onboarding abandonment.

ML Research Intern

August 2023 to September 2023

Compsoft Technologies Pvt. Ltd.

Bangalore, India

- Built a sentiment analysis system using NLP techniques, reaching 90% classification accuracy.
- Developed a Flask web application for real-time inference and deployed it on AWS (EC2, S3) with Docker.
- Applied text preprocessing, feature extraction, and model evaluation across the training pipeline.

Data Science Intern

August 2023 to September 2023

Pantech E-Learning

Chennai, India (Remote)

- Conducted EDA on production datasets and built predictive models (regression, decision trees) in scikit-learn, improving 15% over baseline.
- Contributed to an 87% accuracy COVID-19 prediction model and a 95% accuracy spam classification system; applied deep learning to reach 92% accuracy on churn prediction.

Software Engineer Intern

March 2023 to April 2023

Alltramic

New York, NY (Remote)

- Led development of a Selenium web-automation testing framework, increasing testing speed and accuracy 30%.
- Implemented SMTP email protocols and geo-fencing APIs serving 500+ and 1,000+ users respectively for production application communication.

Open Source Contributor, Layer5 / Meshery (CNCF)

2021 to Present

Cloud Native Computing Foundation

Remote

- Contributed 800+ lines of production Go and Python to Meshery, a CNCF cloud-native service mesh management platform, merging five pull requests through reviews with globally distributed maintainers.
- Followed upstream CNCF contribution workflows and participated in issue triage and community discussions.

PROJECTS

agent-triage AGENTS · OBSERVABILITY

Python, FastAPI, OpenTelemetry, SQLite

- Ranks multi-agent failures by severity, frequency, and recovery from OpenTelemetry or NDJSON traces; ships a CLI, a FastAPI dashboard, an OTLP receiver, and optional LLM root-cause analysis. pip-installable, typed, and tested.

super-mcp-eval

EVALUATION

Python, DuckDB, Streamlit

- Evaluation harness for MCP servers and the agents that use them: schema compliance and tool-selection accuracy with Wilson 95% confidence intervals, DuckDB persistence, and a Streamlit dashboard.

mcp-otel-audit

OBSERVABILITY · GOVERNANCE

Python, OpenTelemetry

- Public, reproducible audit of four MCP OpenTelemetry instrumentations against the official OTel semantic conventions (v1.40.0), with a full report and writeup.

obindoc

RAG · GOVERNANCE

Python, RAG

- Grounded RAG over a PDF with span-level citations, a verifier loop, and a tamper-evident HMAC-chained audit log, in under 900 lines with 29 tests.

klaviyo-agent-demo

AGENTS

Python, LangGraph

- LangGraph multi-agent system that autonomously generates marketing campaigns through tool-calling and orchestration across specialized agents; deployed live.

faramesh-core

GOVERNANCE

Python

- Runtime governance engine for AI agents; contributed roughly 1,900 lines across stacked pull requests to the open-source agent governance layer.

Dungeon Agents

MULTI-AGENT · OBSERVABILITY

Python, LangGraph, Langfuse, Claude

- Multi-agent simulation orchestrating explorer agents on a fog-of-war grid with async inter-agent messaging, a structured NDJSON trace schema, a failure classifier for coordination breakdowns and belief staleness, and an HTML trace viewer.

streamsense

ML RESEARCH

Python, PyTorch, ONNX, FastAPI

- Multimodal human activity recognition on PAMAP2 using a late-fusion CNN plus transformer, with ONNX export, int8 quantization, and a FastAPI inference server for near-real-time inference on wearable IMU streams.

Diplomatic Decoder

NLP · ML

Python, scikit-learn, XGBoost, PyTorch (Bi-LSTM), NLTK

- NLP system analyzing UN diplomatic speeches to predict crisis signals; an XGBoost classifier reached 0.9975 ROC-AUC, cross-validated against a Bidirectional LSTM for sequential pattern recognition.

EDUCATION

M.S. Computer Science	August 2024 to May 2026
Pace University, Seidenberg School of Computer Science	New York, NY
GPA: 3.86 / 4.0. Coursework: Machine Learning, Algorithms, Distributed Systems, Database Systems, Cloud Computing, Natural Language Processing, AI.	
B.E. Computer Science	2020 to 2024
Visvesvaraya Technological University (Jyothy Institute of Technology)	Bangalore, India

CERTIFICATIONS & PUBLICATIONS

Certifications

Anthropic: Building with Claude on Amazon Bedrock (March 2026)
AWS Certified Solutions Architect, Associate (January 2026)
Red Hat Certified System Administrator, RHCSA (July 2022)

Publications

Deep CNN-Based Plant and Medicinal Species Classification, Springer Nature, ICACECS 2023 (corresponding author)
Driver Drowsiness Detection using AI, IJARESM, Vol. 11, November 2023

LEADERSHIP & COMMUNITY

Core Team Member, Tech Club of Jyothy Institute of Technology (2021 to 2024): led the college tech community, organized hackathons and research-focused workshops for 200+ participants annually, and managed national-level event logistics.
Campus Ambassador, SkillVertex (2022): promoted educational programs and grew campus event participation 30%.